Universität Freiburg – Institut für Physik
Applied Theoretical Physics
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Freiburg, 16.6.2025
Due: 23.6.2025

Exercise 7 for '*Computational Physics - Material Science*', SoSe 2025
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Please provide a well documented submission of your solution. Your submission should include

- A pdf file containing the solution to the questions with the corresponding equations that are implemented in your codes. Figures must contain axis titles with corresponding units and a caption.
- The source code should be commented, and the equations given in the pdf file have to be referenced in the source code.
- There is no need to provide the trajectory files.
- In case your code is not working properly, please provide a description of the debugging attempts you did.

Exercise 7.1: Dynamics of a molecular (di-atomic) system

In classical Molecular Dynamics, N di-atomic molecules, $m_i = 1...N$, can be defined as a set of atoms, $i = 1..2N$, that are held together by simple elastic forces. The classical force field replaces the true (quantum) potential through a simplified model Hamiltonian. For dimers (two atoms = di-atoms), a typical expression for the potential part U of the total Hamiltonian ($H = K + U$) is

$$U = U_{\text{bond}} + U_{\text{nonbond}} = \sum_{m=1}^{N} \frac{k_b}{2}(|\vec{b}_m| - b_0)^2 + \sum_{i}^{2N} \sum_{j(m(j)\neq m(i))}^{2N} U_{LJ}(r_{ij})$$

where the first term refers to the intramolecular 'bonded' contribution to the total energy, and the second term serves to describe the intermolecular 'nonbonded' Lennard-Jones (LJ) interactions between all atoms i and j which are not in the same molecule. $\vec{b}_m = \vec{r}_i - \vec{r}_j$ (with $m(i) = m(j)$), k_b and b_0 represent the bond vector of the molecule m, the bond strength, and the equilibrium bond length, respectively. The objective of this exercise is to (i) implement a bonded potential to describe simple molecules, and (ii) calculate the diffusion coefficients of a molecular system.

a) Let us assume a *di-nitrogen* N_2 molecule defined by its equilibrium bond length, $b_0 = 1.07$ Å, and its bond strength, $k_b = 9793$ kcal/mol/nm^2. The mass of a N-atom in a N_2 molecule is 13.97 g/mol. Calculate and plot the time evolution of the bond length, $b(t) = |\vec{b}(t)|$, versus t of a single N_2 molecule with an initial bond length of 1.1 Å and no initial velocity. Do you observe a vibration? What is the corresponding bond frequency ω, and how does it compare with the analytical solution (for a single 3D harmonic oscillator)? 2 points

b) Starting from the implementation of the Lennard-Jones (LJ) fluid in the canonical ensemble using the Berendsen thermostat as developed in Exercise Sheet #5, modify your simulation code to handle a molecular system composed of N di-nitrogen (N_2) molecules. The simulation should include both intermolecular and intramolecular interactions.

Tasks

i) Build the initial configuration using the equilibrium bond length of N_2. Distribute the N molecules on a regular lattice, as was initially done for the LJ fluid. Each diatomic molecule should consist of two atoms connected by a harmonic bond potential. 1 points

ii) Simulate a system of $N = 5^3 = 125$ di-nitrogen molecules in a cubic box of volume V such that the molecular number density is:

$$\rho = \frac{N}{V} = 0.25\,\sigma^{-3}.$$

Run a molecular dynamics simulation at a temperature of $T_d = 300\,\text{K}$ for a total duration of 0.5 nanoseconds using a 1 femtosecond time step. The first 5 picoseconds will be treated as an equilibration phase, and the remaining 495 picoseconds will be used as the production run for data collection and analysis.

During the production run, save the trajectory data to two distinct files: `TrajectoryforC.txt` and `TrajectoryForD.txt`, for the analyses described in parts **c)** and **d)**, respectively.

Only 2 picoseconds of the production run, saved with a frequency of 1 femtosecond, should be written to `TrajectoryforC.txt`. In contrast, the entire production run should be saved to `TrajectoryForD.txt`, using a reduced sampling frequency of 50 femtoseconds.

iii) During the production run, calculate and plot the bond length probability distribution $P(b)$. 1 points

iv) Compute the mean bond length $\langle b(t)\rangle_N$ (averaged over the N atoms) during the production run, and plot it as a function of time. 1 points

v) Calculate and report the average bond energy $\langle U_{\text{bond}}(t)\rangle_N$ over the production run. Does it equilibrate to the expected value? 1 points

vi) Compute and plot the radial distribution function for two different cases: 1 points

(i) r is the distance between the center-of-mass of distinct molecules.

(ii) r is the distance between individual atoms (as for a standard atomic RDF).

vii) Provide an OVITO snapshot of the molecular configuration at the end of the production run. 1 points

c) Each molecule has an associated orientation vector defined by its normalized bond vector:

$$\vec{\Omega}_m(t) = \frac{\vec{b}_m(t)}{|\vec{b}_m(t)|}$$

where $\vec{b}_m(t)$ is the bond vector of molecule m at time t. The following tasks aim at analyzing the rotational dynamics of these molecules as a function of the system density.

Tasks

i) Using the trajectory file `TrajectoryforC.txt` obtained in part **b)**, compute and plot the time-dependent autocorrelation function of molecular orientations given by

$$C(t) = \left\langle \vec{\Omega}(t) \cdot \vec{\Omega}(0) \right\rangle,$$

where the average is taken over all molecules and initial times. This function measures how molecular orientation persists over time. 2 points

ii) Fit the autocorrelation function $C(t)$ to a single-exponential decay model given by

$$C(t) \approx \exp\left(-\frac{t}{\tau_R}\right),$$

and estimate the relaxation time τ_R, which characterizes the timescale over which molecular orientations lose correlation. 1 points

iii) Compare the relaxation time τ_R to the characteristic vibrational timescale of a bond, given by the inverse of the harmonic oscillator frequency:

$$\tau_{\text{bond}} = \omega^{-1}$$

Discuss why τ_R is generally much larger than ω^{-1}, and explain the physical significance of this observation in terms of rotational versus vibrational dynamics. 1 points

iv) Repeat tasks i-ii for each of the following densities: $\rho = 0.5, 0.75$. For each case: 2 points

- Compute and plot the orientation autocorrelation function $C(t)$
- Fit the decay curve and extract the corresponding relaxation time τ_R
- Tabulate and compare the results across densities

v) Analyze how the shape of the autocorrelation function and the relaxation time τ_R depend on system density. Why increasing density is expected to lead to slower orientational relaxation? 1 points

d) Using the trajectory file `TrajectoryforD.txt` obtained in part **b)**, calculate and plot the mean-squared displacement of the molecules' center-of-mass, $\langle \Delta r^2(t) \rangle$ versus t, defined as:

$$\langle \Delta r^2(t) \rangle = \frac{1}{N} \left\langle \sum_m (\mathbf{r}_m(t) - \mathbf{r}_m(t=0))^2 \right\rangle$$

where $\mathbf{r}_m(t)$ is the position vector of the center of mass of molecule m at time t. In three dimensions, the mean-squared displacement $\langle \Delta r^2(t) \rangle$ of a particle undergoing diffusion is related to the diffusion coefficient D by the Einstein relation given by

$$\langle \Delta r^2(t) \rangle = 6Dt.$$

Using the calculated mean-squared displacement, identify the diffusive regime i.e., $\langle \Delta r^2(t) \rangle \propto t$, and report the corresponding diffusion coefficient. 5 points

e) **Bonus:** In the implementation performed in part **b)**, replace the Berendsen thermostat with an Andersen thermostat. Perform the same calculation as in part **b)**, but now for various values of the collision frequency (in fs^{-1}):

$$\nu \in \{0.001,\ 0.01,\ 0.1,\ 1\}$$

For each value of ν, calculate the **mean-squared displacement** (MSD) as described in part **d)**. **Explain** why the diffusion coefficient D depends on the collision frequency ν. In particular, discuss the limiting behaviors for very low and very high ν, and how the Andersen thermostat affects particle momentum and transport properties. 5 points

Numerical values of quantities to be used:

quantity	value (units)
k_B	0.0019849421 (kcal/mol/K)
ϵ	0.0595 (kcal/mol)
σ	0.326 (nm)
T_d	300 (K)
mass	13.97 (g/mol)
b_0	0.107 (nm)
k_b	9793 (kcal/mol/nm^2)
Δt	0.5 (fs)